Counterfactual Conversation Emulation for Medical AI Agents

我们提出 **counterfactual conversation emulation**，用历史医患对话数据 retrospectively 估计 medical AI agent 部署后的效果，替代昂贵缓慢的 prospective clinical trial。

现状的 **gap**：

- Medical AI agent 要进入临床必须做 trial，但 prospective trial 周期几个月、成本上百万，而 agent 每几个月就更新一次——等结果出来，被测的 agent 已经过时。
- 现有 static benchmark（MedQA 等）只测孤立能力，不测部署后真实交互中的 causal effect

以前为什么没人做：

- 药物领域的 trial emulation（Hernán-Robins）假设 treatment 是固定物理干预；agent 的 "treatment"是会随 context 变化的 policy，框架不能直接套用。
- Off-policy evaluation 领域假设 action 是 discrete / low-dimensional；agent 的 action 是 natural language conversation，传统 estimator 失效。
- 这两个领域没有交叉过，medical agent 正好卡在中间。

我们的贡献：第一个把 trial emulation 的 causal identification + OPE 的 estimator 结合，扩展到 conversational action space，让 medical agent 评估从"测准确率"升级到"估 deployment causal effect"

We want to propose **counterfactual conversation emulation** — using historical clinician–patient dialogue data to retrospectively estimate the deployment effect of a medical AI agent, as a scalable alternative to expensive prospective clinical trials.

Why This Matters

The gap in current practice

- Prospective trials of medical AI agents are slow (months) and expensive (often $M+), but agents get updated every few months. By the time a trial finishes, the agent under evaluation is already outdated.
- Static benchmarks (MedQA, USMLE-style) only test isolated capability, not causal effect under deployment — i.e., whether the agent actually changes clinician decisions or patient outcomes in real interactions.

Why no one has done this yet

- **Target trial emulation** (Hernán–Robins) assumes treatment is a fixed physical intervention — but an agent is a *policy* whose behavior depends on context.
- **Off-policy evaluation (OPE)** assumes actions are discrete/low-dimensional — but an agent's action is a *natural-language conversation*. Standard estimators (IPS, DR) fail.

- These two fields have not been bridged. Medical agent evaluation sits exactly in the gap.

Our contribution

The first framework combining target trial emulation's causal identification with OPE's estimators, extended to conversational action spaces — elevating medical agent evaluation from "capability benchmarking" to "deployment causal-effect estimation".

Setting

Input:

- A dataset of historical clinical encounters: $\{(x_i, a_i^{clinician}, y_i)\}$
- x: patient context (symptoms, history)
- a (clinician): actual clinician action — a conversation or decision
- y: outcome (diagnosis quality, management plan, etc.)
- A medical agent policy π(agent) we want to evaluate.

Output:

- An estimate of E[Y | do(A = π(agent)(x))] — the expected outcome if the agent had been deployed instead of (or alongside) the clinician.
- A calibration check: how close is the estimate to ground-truth effect obtained from simulated/real prospective comparison?
- A diagnostic: when does the estimator fail?

Deliverables

4. **Method**: OPE estimator(s) adapted for conversational action spaces.
5. **Benchmark**: Public semi-synthetic eval set with controlled ground truth.
6. **Diagnostic**: Failure-mode analysis + checklist.
7. **Paper draft**: Targeted at ICLR / NeurIPS main track or NeurIPS Datasets & Benchmarks.

Reading List

Tier 1 — Must Read First

Target Trial Emulation (causal inference foundation)

- **Hernán & Robins (2016).** *Using Big Data to Emulate a Target Trial When a Randomized Trial Is Not Available. Am J Epidemiol. →* https://academic.oup.com/aje/article/183/8/758/1739860 Yanbing

- **Hernán, Wang & Leaf (2022).** *Target Trial Emulation: A Framework for Causal Inference From Observational Data. JAMA.* → https://pmc.ncbi.nlm.nih.gov/articles/PMC10400102/
- **Hernán et al. (2025).** *The Target Trial Framework for Causal Inference From Observational Data: Why and When Is It Helpful? Ann Intern Med.* → https://www.acpjournals.org/doi/abs/10.7326/ANNALS-24-01871

Off-Policy Evaluation (methods foundation)

- **Dudík, Langford & Li (2011).** *Doubly Robust Policy Evaluation and Learning. ICML. →* https://arxiv.org/abs/1103.4601 → Xin Wei
- **Dudík, Erhan, Langford & Li (2014).** *Doubly Robust Policy Evaluation and Optimization. Statistical Science.* → https://arxiv.org/abs/1503.02834
- **Wang, Agarwal & Dudík (2017).** *Optimal and Adaptive Off-Policy Evaluation in Contextual Bandits. ICML.* → https://dl.acm.org/doi/10.5555/3305890.3306052

Medical AI Agents (motivating application)

- **Tu et al. (2024).** *Towards Conversational Diagnostic AI (AMIE). Nature. →* https://arxiv.org/abs/2401.05654 → jicheng Liu
- **Saab et al. (2025).** *Multimodal AMIE. Google Research blog.* → https://research.google/blog/amie-gains-vision-a-research-ai-agent-for-multi-modal-diagnostic-dialogue/
- **Brodeur et al. (2026).** *A Prospective Clinical Feasibility Study of a Conversational Diagnostic AI in an Ambulatory Primary Care Clinic (AMIE at BIDMC).* → https://research.google/blog/exploring-the-feasibility-of-conversational-diagnostic-ai-in-a-real-world-clinical-study/

Tier 2 — Agent Benchmarks (for context)

- **Jiang et al. (2025).** *MedAgentBench: A Virtual EHR Environment to Benchmark Medical LLM Agents. NEJM AI.* → https://ai.nejm.org/doi/full/10.1056/AIdbp2500144
- **Schmidgall et al. (2024).** *AgentClinic: a Multimodal Agent Benchmark to Evaluate AI in Simulated Clinical Environments.* → https://arxiv.org/abs/2405.07960 → Xin Wei
- **Yao et al. (2024).** *τ-bench: Tool-Agent-User Benchmark.* → https://sierra.ai/blog/tau-bench-shaping-development-evaluation-agents
- **Ma et al. (2024).** *CliBench: Multifaceted Evaluation of LLMs for Clinical Decision Making.* → https://arxiv.org/abs/2406.09923
- **Kothari & Gupta (2025).** *FHIR-AgentBench: Benchmarking LLM Agents for Realistic EHR QA.* → https://arxiv.org/abs/2509.19319
- **Systematic review (2025).** *Knowledge-Practice Performance Gap in Clinical LLMs: Review of 39 Benchmarks.* → https://pmc.ncbi.nlm.nih.gov/articles/PMC12706444/

Tier 3 — Public Medical Conversation Datasets (candidate data sources)

- **MedDialog (EMNLP 2020).** *0.26M English + 3.4M Chinese patient–doctor conversations.* → https://github.com/UCSD-AI4H/Medical-Dialogue-System
- **MTS-Dialog (EACL 2023).** *~1700 dialogue–note snippet pairs.* → https://github.com/abachaa/MTS-Dialog
- **ACI-Bench (2023).** *207 full role-played encounter dialogue–note pairs.* → https://github.com/microsoft/clinical_visit_note_summarization_corpus
- **PriMock57 (2022).** *57 mock primary-care consultations (audio + transcript).* → https://arxiv.org/abs/2204.00333
- **NoteChat (2023).** *Synthetic doctor–patient dialogues from clinical notes.* → https://huggingface.co/datasets/akemiH/NoteChat
- **OSCE Respiratory (Scientific Data 2022).** *Simulated OSCE-format respiratory cases.* → https://www.nature.com/articles/s41597-022-01423-1
- **AnnoMI.** *Motivational interviewing, expert-annotated.* → https://github.com/uccollab/AnnoMI
- **MIMIC-III / IV + Notes**. ICU, no conversation but rich context. Requires PhysioNet credentialing.

Tier 4 — Background & Extensions

Counterfactual reasoning in RL / contextual bandits

- **Jiang & Li (2016).** *Doubly Robust Off-Policy Evaluation for Reinforcement Learning.* → https://arxiv.org/abs/1511.03722 Jicheng
- **Off-Policy Evaluation for Large Action Spaces via Embeddings https://arxiv.org/pdf/2202.06317 Yanbing**
- **Towards Automatic Evaluation of Dialog Systems: A Model-Free Off-Policy Evaluation Approach https://aclanthology.org/2021.emnlp-main.589.pdf Xin Wei**
- **Thomas & Brunskill (2016).** *Data-Efficient Off-Policy Policy Evaluation for Reinforcement Learning. ICML.* → https://arxiv.org/abs/1604.00923
- **Li (2019).** *A Perspective on Off-Policy Evaluation in Reinforcement Learning.* → https://lihongli.github.io/papers/li19perspective.pdf

Clinical AI evaluation frameworks (regulatory context)

- **DECIDE-AI, CONSORT-AI, TRIPOD+AI guidelines**. Reporting standards for clinical AI studies. See EQUATOR Network: https://www.equator-network.org/